



CES_SALCON-SDC JV

TUAS WATER RECLAMATION PLANT
PROJECT TITLE: CONTRACT 4B – DIGESTION PRE-TREATMENT AND
SIDESTREAM NITROGEN REMOVAL FACILITIES

CLIENT:



CONSULTANT:



SUBMISSION INFORMATION

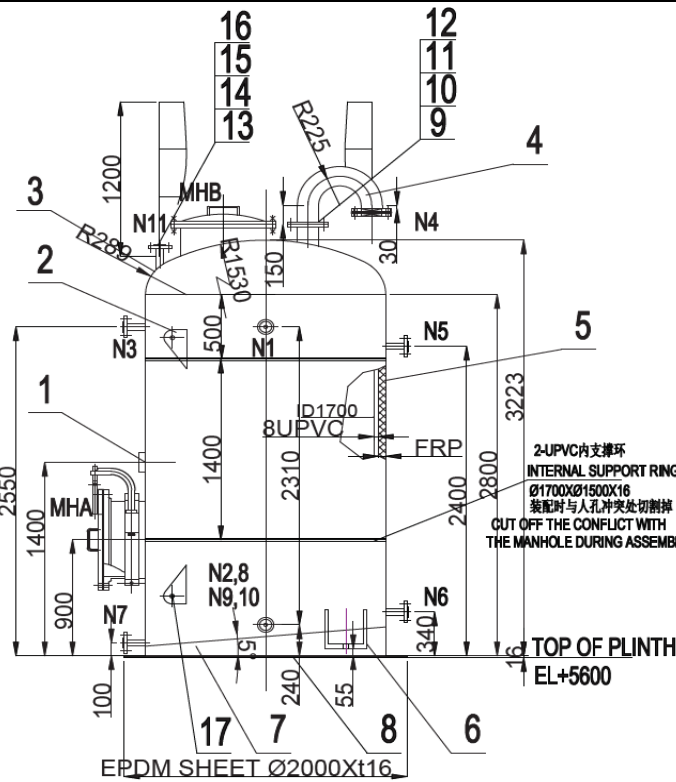
Document Title:	DATA SHEET FOR FERRIC CHLORIDE STORAGE TANK 1/2		
Document No.:	131242-WST02-DS-M-5401 (Previously issued as 131242-WST02-DS-M-0001 R0)		
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	Name / Company	Designation	Signature
Prepared by	David Li/ Lu Yibo	Mech Engr/ Mech. Engr Mgr	
Checked by	Franklin Song	Design Manager	
Approved by	Keith Lim	Project Manager	

Document Revision History Record

Document Number : 131242-WST02-DS-M-5401
(Previously issued as 131242-WST02-DS-M-0001 R0)
Document Title : DATA SHEET FOR FERRIC CHLORIDE STORAGE TANK
1/2

REV	STATUS	DATE ISSUED	UPDATE/AMENDMENT DETAILS
1	IFA	20-Sep-2024	Issued for Approval
2	IFA	25-Mar-2025	Issued for Approval

FRP TANK DATA SHEET			DOC. No. 131242-WST02-DS-M-5401		REV. NO 2		SHEET No. 1 of 2		REVISION
EQ. NAME FERRIC CHLORIDE STORAGE TANK 1 / 2			TAG No. WST02-401-TNK-0101 / TNK-0201						
PROJECT NAME CONTRACT 4B - DIGESTION PRE-TREATMENT AND SIDESTREAM NITROGEN REMOVAL FACILITIES			Client CES_SALCON-SDC JV						
Contract No. 1230207			End User PUBLIC UTILITIES BOARD (PUB)						
1.00 MANUFACTURER									
1.01 TANK TYPE / MODEL			VERTICAL FRP TANK						
1.02 MANUFACTURER			JIZHOU ZHONGYI FRP CO., LTD						
1.03 COUNTRY OF ORIGIN			CHINA						
2.00 OPERATING AND DESIGN									
			UNIT		DATA				
2.01 OPERATING FLUID DESCRIPTION			Ferric Chloride (41%)						
2.02 FLUID DENSITY AT DESIGN			kg/m ³		1,400				
2.03 DESIGN CODE			BS EN 13121-3:2008 + Amendment 1: 2010 and BS EN 13923:2005						
2.04 CODE STAMP REQUIRED			☐ Yes ☒ No						
2.05 DESIGN PRESSURE @ TEMPERATURE			Full Process Liquid + 150mm above top of tank @ 60 °C						
2.06 OPERATING PRESSURE @ TEMPERATURE - INTERNAL			Atmospheric @ 30 °C						
2.07 HYDROSTATIC TEST PRESSURE			mmWATER		5420				
2.08 CORROSION ALLOWANCE			mm		N.A.				
2.09 CAPACITY (NET VOLUME / GROSS VOLUME)			m ³		4 / 5.1				
2.16 VESSEL DIAMETER (INSIDE)			mm		1700				
2.17 VESSEL STRAIGHT SHELL HEIGHT (BOTTOM to TOP HEAD TL)			mm		2800				
2.18 HEAD TYPE			2:1 Elliptical						
2.19 EMPTY WEIGHT			kg		1400				
2.20 OPERATING WEIGHT			kg		8540				
2.21 FREE BOARD (HH to Top of Head)			mm		1023				
3.00 ADDITIONAL DESIGN CONDITIONS									
			DATA						
3.01 WIND LOAD (BASIC VELOCITY / CODE)			20 m/s SS EN 1991-1-4 : 2009						
3.02 SEISMIC			N.A.						
4.00 MATERIALS									
			DATA						
4.01 SHELL			uPVC Lined FRP						
4.02 HEADS			uPVC Lined FRP						
4.03 BASE			uPVC Lined FRP						
4.04 NOZZLE NECKS / NOZZLE FLANGES			uPVC Lined FRP						
4.05 EXTERNAL BOLTING (Nozzle Bolt, Hold Down Bolt)			SS316						
4.06 GASKETS			EPDM						
4.07 EXTERNAL ATTACHMENTS (eg. Lifting Lug, Support Clip)			Stainless Steel SS304 with FRP Strapping						
4.08 NAMEPLATE			SS316						
4.09 HAND RAIL			Carbon Steel + Coating as per PUB specification G1.5 and G4.61.2 (e)/(vii)						
5.00 PAINTING/COATING/FINISHES									
			DATA						
5.01 RESIN			Vinyl Ester as per PUB specification G4.47.5(q)(viii) and G4.61.2(b)						
5.02 EXTERNAL COATING			Ultraviolet absorbers/inhibitor shall be add to surfacing resin/exterior gel coat						
5.02 EXTERNAL COLOR			RAL 6019 (PASTEL GREEN)						
			Equivalent to BS4800 14C31						

FRP TANK DATA SHEET					DOC. No.	REV. NO	SHEET No.
					131242-WST02-DS-M-5401	2	2 of 2
EQ. NAME					TAG No.		
FERRIC CHLORIDE STORAGE TANK 1 / 2					WST02-401-TNK-0101 / TNK-0201		
6.00	NOZZLE SCHEDULE						
MARK	SIZE / DN	RATING	PROJECTION	TYPE	SERVICE / DESCRIPTION		
N1	25	150#	150	ASME B16.5 Flat Face	Level Gauge		
N2	25	150#	150	ASME B16.5 Flat Face	Level Gauge		
N3	50	150#	150	ASME B16.5 Flat Face	Inlet		
N4	150	150#	150	ASME B16.5 Flat Face	Vent (with gooseneck and insect screen)		
N5	100	150#	150	ASME B16.5 Flat Face	Overflow		
N6	50	150#	150	ASME B16.5 Flat Face	Outlet		
N7	50	150#	150	ASME B16.5 Flat Face	Drain		
N8	25	150#	150	ASME B16.5 Flat Face	Level Transmitter		
N9	25	150#	150	ASME B16.5 Flat Face	Level Switch High		
N10	25	150#	150	ASME B16.5 Flat Face	Level Switch Low		
N11	50	150#	150	ASME B16.5 Flat Face	Spare with Blind Flange, Gasket and Fastening		
MH-A	600	150#	per drawing	ASME B16.5 Flat Face	Manhole		
MH-B	600	150#	per drawing	ASME B16.5 Flat Face	Manhole		
7.00	SKETCH						
 LAMINATION SCHEDULE铺层表: S---30g/m³ SURFACE VEIL表面毡 δ=0.31 G---250g/m³ TRANSITION RESIN过渡树脂 A410-901# δ=0.217 M---450g/m³ CHOPPED STRAND MAT短切毡 δ=1.09 R---600g/m³ WOVEN ROVING玻璃布 δ=0.76 H---651g/m³ WINDING ROVING缠绕纱 90° δ=0.5 U---430g/m³ UNIDIRECTIONAL CLOTH单向布 δ=0.42 P---300g/m³ SPRAY ROVING喷射纱 δ=0.65 LINER: UPVC 1LAYER1层 8mm TOP HEAD封头:UPVC, G, M, 3 (MR), M, S (8+8.25mm) TANK SHELL罐体:UPVC, G, M, 2 (P, 2H, 2U), 2H, S (8+7.6m) SLOPE BOTTOM封底:UPVC, G, 6 (M, R), M, S (8+12.7mm) 2-UPVC内支撑环 INTERNAL SUPPORT RING Ø1700XØ1500X16 装配时与人孔冲突处切割掉 CUT OFF THE CONFLICT WITH THE MANHOLE DURING ASSEMBL TOP OF PLINTH EL+5600 EPDM SHEET Ø2000X16							
8.00	NOTES						
8.01	Nozzle shall finish flush with inside surface of tank, unless otherwise indicated.						
8.02	Flanged nozzles shall be fabricated of the same material as the tank, all nozzle (except manhole) shall be gusseted to the tank.						
8.03	The vent and overflow pipes shall be properly supported and arranged so that they can be dismantled for inspection.						
8.04	Tank roofs shall be provided with a nonslip finish over the entry roof surface, top head including the top manhole cover shall be designed for personnel loads.						
8.05	Additional reinforcement shall be provided as necessary to support the required accessories.						
8.06	External clips if applicable shall be provided by Supplier.						
8.07	Tank shall be mounted above flood level on flat concrete plinths and cushioned using compressible and corrosion resistant EPDM sheet of not less than 16 mm thick.						
8.08	The cushion sheet shall have length and width at least 150 mm greater than the tank diameter underneath the tank.						
8.09	Supplier shall provide Recommended Bolt Tightening Torques values for all boltings of nozzles and anchors.						
REV	DATE	BY	CHKD	APPR	AUTH	REVISION DESCRIPTION	
0	13 09 2024	DL	YB	FL	KL	Issue for Approval	
1	25 03 2025	DL	YB	FL	KL	Issue for Approval	

APPENDIX

MSDS AND CHEMICAL COMPATIBILITY



化学品

Maximum Temperature

最高推荐温度, (°C/°F)

浓度(%)	SWANCOR 901	SWANCOR 901-N	SWANCOR 905	SWANCOR 905-N	SWANCOR 907	SWANCOR 980	SWANCOR 977-S
氯化钠 pH < 2.5, Cl2 Sat'd (见氯化盐水) Sodium Chloride, pH < 2.5, Cl2 Sat'd (see Chlorinated Brine)							
氯化钠 (盐水, 盐) Sodium Chloride (Brine, Salt)	All	100/210	100/210	100/210	100/210	80/180	—
氯化钠 / 氯酸钠 Sodium Chloride/Sodium Chlorate	20/34	100/210	100/210	100/210	100/210	—	—
氯化钠 / 氢氧化钠 [1,4] Sodium Chloride/Sodium Hydroxide [1,4]	0.5-10/0.1-2	80/180	60/140	70/160	40/100	40/100	—
氯化钠 / 盐酸 [3,4,8] Sodium Chloride/Hydrochloric Acid [3,4,8]	Sat'd. NaCl/5	100/210	—	100/210	—	—	—
氯化钠 / 氧化镁 / 石灰 Sodium Chloride/Magnesium Oxide/Lime	0.5-26/0.1-20/0.1-10	100/210	100/210	100/210	100/210	80/180	—
氯化钠 / 乙基香兰素 Sodium Chloride/Ethyl Vanillin	0.1-25/1	50/120	50/120	—	—	—	—
氯化钠与氯, pH > 9 (见氯化盐水) Sodium Chloride with Chlorine, pH > 9 (see Chlorinated Brine)							
氯化钠与氯, pH 2.5-9 [9] Sodium Chloride with Chlorine, pH 2.5-9 [9]	All	LS	—	LS	LS	LS	—
氯化镍 Nickel Chloride	All	100/210	100/210	100/210	100/210	80/180	—
氯化氢, 干气 Hydrogen Chloride, Dry Gas	100	100/210	175/350	100/210	175/350	80/180	195/380
氯化氢, 湿气 Chelant Hydrogen Chloride, Wet Gas	100	100/210	110/230	100/210	110/230	80/180	—
氯化清洗机 (烟草和通风系统) Chlorination Washer (Hoods & Vent Systems)	Fumes	80/180	95/200	80/180	95/200	95/200	65/150
氯化溶剂回收 (见特定溶剂) Chlorinated Solvent Recovery (see Specific Solvents)							
氯化石蜡 Chlorinated Paraffin Wax	100	80/180	—	80/180	—	—	—
氯化铁 / 盐酸 [3,6,8] Ferric Chloride/Hydrochloric Acid [3,6,8]	0-29/1-20	80/180	105/220	80/180	105/220	80/180	—
氯化铁 Ferric Chloride	All	100/210	100/210	100/210	100/210	80/180	—
氯化铜 Copper Chloride	All	100/210	100/210	100/210	100/210	80/180	—
氯化烷基 (C8-C18) Alkyl (C8-C18) Chloride	All	80/180	95/200	95/200	100/210	—	—
氯化锡 Stannic Chloride	All	100/210	100/210	100/210	100/210	80/180	—
氯化锌 Zinc Chloride	All	100/210	100/210	—	100/210	80/180	—
氯化亚砷 Thionyl Chloride	100	NR	NR	NR	NR	LS	NR
氯化亚铁、氯化锰、氯化铁 / 盐酸 [3,6,8] Ferrous Chloride, Manganese Chloride, Ferric Chloride/Hydrochloric Acid [3,6,8]	1-60/0-20	80/180	100/210	100/210	100/210	80/180	—
氯化亚铁 / 盐酸 [3,6,8] Ferrous Chloride/Hydrochloric Acid [3,6,8]	0-29/1-20	80/180	100/210	80/180	100/210	80/180	—
氯化亚铁 Ferrous Chloride	All	100/210	100/210	100/210	100/210	80/180	—
氯化亚锡 Stannous Chloride	All	100/210	100/210	100/210	100/210	80/180	—
氯化盐水 (见氯化盐水) Chlorinated Brine (see Brine, Chlorinated)							
氯化盐水, pH < 2.5 [8] Brine, chlorinated, pH < 2.5 [8]	All	80/180	80/180	80/180	95/200	95/200	—
氯化盐水, pH > 9 [2,4,8] Brine, chlorinated, pH > 9 [2,4,8]	All	80/180	80/180	80/180	65/150	65/150	—
氯化盐水, pH 2.5-9 Brine, chlorinated, pH 2.5-9	All	40/100	40/100	40/100	40/100	40/100	—
氯化纸浆 Chlorinated Pulp	All	80/180	90/195	90/195	95/200	95/200	—
氯磺酸 Chlorosulfonic Acid	10	NR	NR	NR	NR	NR	—
氯磺酸 Chlorosulfonic Acid	100	NR	NR	NR	NR	NR	—
氯甲苯 Chlorotoluene	100	25/80	40/100	25/80	40/100	40/100	NR

Chemical Resistance Tool

Chemical resistance: How to find the right material for aggressive media

Not every pipe material is suitable for transporting aggressive liquids and gases. The use of incorrect materials can lead to problems. With this online tool, you can now find the right materials for your project both quickly and easily.

- + Select the medium you intend to transport.
- + ChemRes PLUS will then show you suitable materials in a clearly arranged table.

The listed ratings are valid for conditions listed and assumed to be for continual exposure to medium at temperature listed and for use in a pressure piping system. However, while a negative or conditional rating may be noted, a suitable application of a particular material/medium combination may be given under different process conditions (e.g. drainage systems, temporary exposure, briefly exceeding maximum temperature, etc.). Please contact GF for more information.

Media List

Materials

Units

2 selected



17 selected



°C / bar



Iron chlorides, aqueous solutions

Formula: FeCl₃, FeCl₂

Synonyme: ferric Chloride, ferrous chloride

Comments: Tangit Rapid is ok for common Ferric chloride concentrations for water treatment.

-	not applicable	0	individual evaluation, please contact GF	+	(AF)	conditionally applicable	++	unconditionally applicable
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Piping Systems

	20 °C	40 °C	60 °C	80 °C	100 °C	120 °C	140 °C
--	-------	-------	-------	-------	--------	--------	--------

PVC-U

++	++	++	-			
----	----	----	---	--	--	--

PVC-C

++	++	++	++	-		
----	----	----	----	---	--	--

ABS

0						
---	--	--	--	--	--	--

PE

++	++	++	-			
----	----	----	---	--	--	--

PP

++	++	++	++	-		
----	----	----	----	---	--	--

PVDF

++	++	++	++	++	0	
----	----	----	----	----	---	--

ECTFE

Sealing materials

	20 °C	40 °C	60 °C	80 °C	100 °C	120 °C	140 °C
--	-------	-------	-------	-------	--------	--------	--------

EPDM

++	++	++	++	0		
----	----	----	----	---	--	--

FKM

++	++	++	++	0		
----	----	----	----	---	--	--

NBR

++	++	++	++	0		
----	----	----	----	---	--	--

PTFE / PFA /

FEP

Jointing of PVC

	20 °C	40 °C	60 °C	80 °C	100 °C	120 °C	140 °C
--	-------	-------	-------	-------	--------	--------	--------

Tangit PVC-U

++	++	++	-			
----	----	----	---	--	--	--

Tangit PVC-C

++	++	++	++	-		
----	----	----	----	---	--	--

Plus

Tangit DTX

++	++	++	-			
----	----	----	---	--	--	--

Tangit RAPID	++	++	0	-	
Tangit PVC-U Plus	++	++	++	-	
IR-welding	++	++	++	-	

The ChemRes Plus database is for general information only, without accepting any warranty, representation or liability. Caution: In order to cope with different demands for extrusion (pipes) and injection molding (fittings, valves) the compositions of both resins may be slightly different. That in individual cases may give rise to slight differences in chemical resistance behaviour. Please note: The Information provided by the ChemRes Plus database does not reflect relevant parameters, particularly but not limited to long-term and dynamic effects, nature of the chemical attack, mixture effects, etc. In case of any queries we recommend to contact GF (chem@georgfischer.com). GF will be happy to share with you its extensive practical experience, gained over decades in the field of polymer piping system applications in industry and chemical engineering.

Here’s what the results mean:

- ++ (Green background): Fully recommended, no influence on material properties.
- + (Blue background): Recommended together with the specified reduction factor. Influence on material properties, e.g. by penetration or swelling, is possible, but no chemical attack.
- 0 (Yellow background): Restricted recommendation. From this temperature, recommendations are not generally given, only on the basis of an individual assessment. Please contact GF Piping Systems directly.
- (Red background): Not suitable. Incompatibility of material and medium.

Contact

 **George Fischer Pte. Ltd, Vietnam Representative Office**
106 Nguyễn Văn Trỗi
700000 Phú Nhuận
Vietnam



Safety data sheet Safety data sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)

FOR REFERENCE
ACTUAL SDS SHALL BE BASED ON
CHEMICAL MANUFACTURER SELECTED



Iron(III) chloride solution 40 %

article number: **7750**

Version: **4.0 en**

Replaces version of: 2022-01-12

Version: (3)

date of compilation: 2016-12-01

Revision: 2023-03-09

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Identification of the substance

Iron(III) chloride solution 40 %

Article number

7750

1.2 Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses:

Laboratory chemical
Laboratory and analytical use

Uses advised against:

Do not use for products which come into contact with foodstuffs. Do not use for private purposes (household).

1.3 Details of the supplier of the safety data sheet

Carl Roth GmbH + Co KG
Schoemperlenstr. 3-5
D-76185 Karlsruhe
Germany

Telephone: +49 (0) 721 - 56 06 0

Telefax: +49 (0) 721 - 56 06 149

e-mail: sicherheit@carlroth.de

Website: www.carlroth.de

Competent person responsible for the safety data sheet: Department Health, Safety and Environment

e-mail (competent person):

sicherheit@carlroth.de

1.4 Emergency telephone number

Name	Street	Postal code/city	Telephone	Website
National Poisons Information Service City Hospital	Dudley Rd	B187QH Birmingham	844 892 0111	

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification acc. to GHS

Section	Hazard class	Cat-egory	Hazard class and category	Hazard statement
2.16	Substance or mixture corrosive to metals	1	Met. Corr. 1	H290
3.10	Acute toxicity (oral)	4	Acute Tox. 4	H302
3.2	Skin corrosion/irritation	2	Skin Irrit. 2	H315
3.3	Serious eye damage/eye irritation	1	Eye Dam. 1	H318
3.4S	Skin sensitisation	1	Skin Sens. 1	H317

Safety data sheet Safety data sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Iron(III) chloride solution 40 %

article number: 7750

For full text of abbreviations: see SECTION 16

2.2 Label elements

Labelling

Signal word

Danger

Pictograms

GHS05, GHS07



Hazard statements

H290	May be corrosive to metals
H302	Harmful if swallowed
H315	Causes skin irritation
H317	May cause an allergic skin reaction
H318	Causes serious eye damage

Precautionary statements

Precautionary statements - prevention

P280	Wear protective gloves/eye protection
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Precautionary statements - response

P302+P352	IF ON SKIN: Wash with plenty of water
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310	Immediately call a POISON CENTER/doctor

Hazardous ingredients for labelling: Iron(III) chloride, Hydrochloric acid %

2.3 Other hazards

Results of PBT and vPvB assessment

Does not contain a PBT-/vPvB-substance in a concentration of $\geq 0,1\%$.

Endocrine disrupting properties

Does not contain an endocrine disruptor (EDC) in a concentration of $\geq 0,1\%$.

SECTION 3: Composition/information on ingredients

3.1 Substances

not relevant (mixture)

3.2 Mixtures

Description of the mixture

Safety data sheet Safety data sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Iron(III) chloride solution 40 %

article number: 7750

Name of sub-stance	Identifier	Wt%	Classification acc. to GHS	Pictograms	Notes
Iron(III) chloride	CAS No 7705-08-0 EC No 231-729-4	39 – 41	Met. Corr. 1 / H290 Acute Tox. 4 / H302 Skin Irrit. 2 / H315 Eye Dam. 1 / H318 Skin Sens. 1 / H317	 	
Hydrochloric acid %	CAS No 7647-01-0 EC No 231-595-7 Index No 017-002-01-X	≤ 1	Met. Corr. 1 / H290 Skin Corr. 1B / H314 Eye Dam. 1 / H318 STOT SE 3 / H335	 	B(a) GHS-HC IOELV

Notes

B(a): The classification refers to an aqueous solution

GHS-HC: Harmonised classification (the classification of the substance corresponds to the entry in the list according to 1272/2008/EC, Annex VI)

IOELV: Substance with a community indicative occupational exposure limit value

Name of sub-stance	Identifier	Specific Conc. Limits	M-Factors	ATE	Exposure route
Iron(III) chloride	CAS No 7705-08-0 EC No 231-729-4	-	-	500 mg/kg	oral
Hydrochloric acid %	CAS No 7647-01-0 EC No 231-595-7	Met. Corr. 1; H290: C ≥ 0,1 % Skin Corr. 1B; H314: C ≥ 25 % Skin Irrit. 2; H315: 10 % ≤ C < 25 % Eye Dam. 1; H318: C ≥ 25 % Eye Irrit. 2; H319: 10 % ≤ C < 25 % STOT SE 3; H335: C ≥ 10 %	-	-	

For full text of abbreviations: see SECTION 16

SECTION 4: First aid measures

4.1 Description of first aid measures



General notes

Take off immediately all contaminated clothing.

Following inhalation

Provide fresh air. In all cases of doubt, or when symptoms persist, seek medical advice.

Following skin contact

Rinse skin with water/shower. After contact with skin, wash immediately with plenty of water. In case of skin reactions, consult a physician. In case of skin irritation, consult a physician.

Following eye contact

In case of contact with eyes flush immediately with plenty of flowing water for 10 to 15 minutes holding eyelids apart and consult an ophthalmologist.

Safety data sheet Safety data sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Iron(III) chloride solution 40 %

article number: 7750

Following ingestion

Rinse mouth immediately and drink plenty of water. Rinse mouth with water (only if the person is conscious). Call a physician immediately.

4.2 Most important symptoms and effects, both acute and delayed

Irritation, Allergic reactions, Vomiting, Corrosion, Gastric perforation, Risk of serious damage to eyes, Risk of blindness

4.3 Indication of any immediate medical attention and special treatment needed

none

SECTION 5: Firefighting measures

5.1 Extinguishing media



Suitable extinguishing media

co-ordinate firefighting measures to the fire surroundings
water spray, alcohol resistant foam, dry extinguishing powder, BC-powder, carbon dioxide (CO₂)

Unsuitable extinguishing media

water jet

5.2 Special hazards arising from the substance or mixture

Non-combustible.

5.3 Advice for firefighters

In case of fire and/or explosion do not breathe fumes. Fight fire with normal precautions from a reasonable distance. Wear self-contained breathing apparatus. Wear full chemical protective clothing.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures



For non-emergency personnel

Use personal protective equipment as required. Avoid contact with skin, eyes and clothes. Do not breathe vapour/spray.

6.2 Environmental precautions

Keep away from drains, surface and ground water. The product is an acid. Before discharge into sewage plants the product normally needs to be neutralised.

6.3 Methods and material for containment and cleaning up

Advice on how to contain a spill

Covering of drains.

Advice on how to clean up a spill

Absorb with liquid-binding material (sand, diatomaceous earth, acid- or universal binding agents).

Safety data sheet Safety data sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Iron(III) chloride solution 40 %

article number: 7750

Other information relating to spills and releases

Place in appropriate containers for disposal. Ventilate affected area.

6.4 Reference to other sections

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Handle and open container with care.

Advice on general occupational hygiene

Wash hands before breaks and after work. Keep away from food, drink and animal feedingstuffs.

7.2 Conditions for safe storage, including any incompatibilities

Keep container tightly closed. Keep only in original container.

Incompatible substances or mixtures

Observe hints for combined storage.

Consideration of other advice:

Specific designs for storage rooms or vessels

Recommended storage temperature: 15 – 25 °C

7.3 Specific end use(s)

No information available.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

National limit values

Occupational exposure limit values (Workplace Exposure Limits)

Cou ntr y	Name of agent	CAS No	Identi- fier	TW A [pp m]	TWA [mg/ m ³]	STE L [pp m]	STEL [mg/ m ³]	Ceil ing- C [pp m]	Ceil- ing-C [mg/ m ³]	Nota- tion	Source
EU	hydrogen chloride	7647-01-0	IOELV	5	8	10	15				2000/39/ EC
GB	hydrogen chloride	7647-01-0	WEL	1	2	5	8			ga	EH40/ 2005

Notation

Ceiling-C Ceiling value is a limit value above which exposure should not occur

ga As gases and aerosols

STEL Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15-minute period (unless otherwise specified)

TWA Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours time-weighted average (unless otherwise specified)

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Relevant DNELs of components of the mixture						
Name of substance	CAS No	End-point	Threshold level	Protection goal, route of exposure	Used in	Exposure time
Iron(III) chloride	7705-08-0	DNEL	2,8 mg/kg bw/day	human, dermal	worker (industry)	chronic - systemic effects
Hydrochloric acid %	7647-01-0	DNEL	8 mg/m ³	human, inhalatory	worker (industry)	chronic - local effects
Hydrochloric acid %	7647-01-0	DNEL	15 mg/m ³	human, inhalatory	worker (industry)	acute - local effects

8.2 Exposure controls

Individual protection measures (personal protective equipment)

Eye/face protection



Use safety goggle with side protection.

Skin protection



• hand protection

Wear suitable gloves. Chemical protection gloves are suitable, which are tested according to EN 374. For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves. The times are approximate values from measurements at 22 ° C and permanent contact. Increased temperatures due to heated substances, body heat etc. and a reduction of the effective layer thickness by stretching can lead to a considerable reduction of the breakthrough time. If in doubt, contact manufacturer. At an approx. 1.5 times larger / smaller layer thickness, the respective breakthrough time is doubled / halved. The data apply only to the pure substance. When transferred to substance mixtures, they may only be considered as a guide.

• type of material

NBR (Nitrile rubber)

• material thickness

≥0,3 mm

• breakthrough times of the glove material

>480 minutes (permeation: level 6)

• other protection measures

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

Respiratory protection

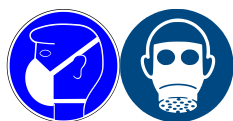
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Respiratory protection necessary at: Aerosol or mist formation. Type: B-P2 (combined filters for acidic gases and particles, colour code: Grey/White).

Environmental exposure controls

Keep away from drains, surface and ground water.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	dark brown
Odour	stinging
Melting point/freezing point	-12 °C
Boiling point or initial boiling point and boiling range	not determined
Flammability	non-combustible
Lower and upper explosion limit	not determined
Flash point	not determined
Auto-ignition temperature	not determined
Decomposition temperature	not relevant
pH (value)	<1 (in aqueous solution: 100 mg/cm ³ , 20 °C)
Kinematic viscosity	6,993 mm ² /s at 20 °C
Dynamic viscosity	10 mPa s at 20 °C
<u>Solubility(ies)</u>	
Water solubility	miscible in any proportion
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	not relevant (inorganic)
Vapour pressure	not determined
<u>Density and/or relative density</u>	
Density	1,43 g/cm ³ at 20 °C
Relative vapour density	information on this property is not available
Particle characteristics	not relevant (liquid)

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Other safety parameters

Oxidising properties none

9.2 Other information

Information with regard to physical hazard classes:

Corrosive to metals category 1: corrosive to metals

Other safety characteristics:

Miscibility completely miscible with water

SECTION 10: Stability and reactivity

10.1 Reactivity

Substance or mixture corrosive to metals.

10.2 Chemical stability

The material is stable under normal ambient and anticipated storage and handling conditions of temperature and pressure.

10.3 Possibility of hazardous reactions

Violent reaction with: Alkali (lye), Metals

10.4 Conditions to avoid

Keep away from heat.

10.5 Incompatible materials

different metals

Release of flammable materials with

Metals, Light metals (due to the release of hydrogen in an acid/alkaline medium)

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Test data are not available for the complete mixture.

Classification procedure

The method for classification of the mixture is based on ingredients of the mixture (additivity formula).

Classification acc. to GHS

Acute toxicity

Harmful if swallowed.

Acute toxicity estimate (ATE) of components of the mixture

Name of substance	CAS No	Exposure route	ATE
Iron(III) chloride	7705-08-0	oral	500 mg/kg

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Acute toxicity of components of the mixture

Name of substance	CAS No	Exposure route	Endpoint	Value	Species
Iron(III) chloride	7705-08-0	oral	LD50	500 mg/kg	rat
Iron(III) chloride	7705-08-0	dermal	LD50	>2.000 mg/kg	rat

Skin corrosion/irritation

Causes skin irritation.

Serious eye damage/eye irritation

Causes serious eye damage.

Respiratory or skin sensitisation

May cause an allergic skin reaction.

Germ cell mutagenicity

Shall not be classified as germ cell mutagenic.

Carcinogenicity

Shall not be classified as carcinogenic.

Reproductive toxicity

Shall not be classified as a reproductive toxicant.

Specific target organ toxicity - single exposure

Shall not be classified as a specific target organ toxicant (single exposure).

Specific target organ toxicity - repeated exposure

Shall not be classified as a specific target organ toxicant (repeated exposure).

Aspiration hazard

Shall not be classified as presenting an aspiration hazard.

Symptoms related to the physical, chemical and toxicological characteristics

• If swallowed

vomiting, nausea, Liver and kidney damage

• If in eyes

Causes serious eye damage, risk of blindness

• If inhaled

irritant effects

• If on skin

causes skin irritation, May produce an allergic reaction, pruritis, localised redness

• Other information

none

11.2 Endocrine disrupting properties

Does not contain an endocrine disruptor (EDC) in a concentration of $\geq 0,1\%$.

11.3 Information on other hazards

There is no additional information.

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SECTION 12: Ecological information

12.1 Toxicity

Shall not be classified as hazardous to the aquatic environment.

12.2 Persistence and degradability

Data are not available.

12.3 Bioaccumulative potential

Data are not available.

Bioaccumulative potential of components of the mixture

Name of substance	CAS No	BCF	Log KOW	BOD5/COD
Iron(III) chloride	7705-08-0		-4 (24 °C)	

12.4 Mobility in soil

Data are not available.

12.5 Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB. Does not contain a PBT-/vPvB-substance in a concentration of $\geq 0,1\%$.

12.6 Endocrine disrupting properties

Does not contain an endocrine disruptor (EDC) in a concentration of $\geq 0,1\%$.

12.7 Other adverse effects

Data are not available.

SECTION 13: Disposal considerations

13.1 Waste treatment methods



This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.

Sewage disposal-relevant information

Do not empty into drains.

Waste treatment of containers/packagings

It is a dangerous waste; only packagings which are approved (e.g. acc. to ADR) may be used.

13.2 Relevant provisions relating to waste

The allocation of waste identity numbers/waste descriptions must be carried out according to the EEC, specific to the industry and process.

Properties of waste which render it hazardous

HP 4 irritant - skin irritation and eye damage

HP 6 acute toxicity

HP 13 sensitising

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13.3 Remarks

Waste shall be separated into the categories that can be handled separately by the local or national waste management facilities. Please consider the relevant national or regional provisions.

SECTION 14: Transport information

14.1 UN number or ID number

ADRRID	UN 2582
IMDG-Code	UN 2582
ICAO-TI	UN 2582

14.2 UN proper shipping name

ADRRID	FERRIC CHLORIDE SOLUTION
IMDG-Code	FERRIC CHLORIDE SOLUTION
ICAO-TI	Ferric chloride solution

14.3 Transport hazard class(es)

ADRRID	8
IMDG-Code	8
ICAO-TI	8

14.4 Packing group

ADRRID	III
IMDG-Code	III
ICAO-TI	III

14.5 Environmental hazards

non-environmentally hazardous acc. to the dangerous goods regulations

14.6 Special precautions for user

Provisions for dangerous goods (ADR) should be complied within the premises.

14.7 Maritime transport in bulk according to IMO instruments

The cargo is not intended to be carried in bulk.

14.8 Information for each of the UN Model Regulations

Agreement concerning the International Carriage of Dangerous Goods by Road (ADR) Additional information

Proper shipping name	FERRIC CHLORIDE SOLUTION
Particulars in the transport document	UN2582, FERRIC CHLORIDE SOLUTION, 8, III, (E)
Classification code	C1
Danger label(s)	8



Excepted quantities (EQ)	E1
Limited quantities (LQ)	5 L

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Transport category (TC)	3
Tunnel restriction code (TRC)	E
Hazard identification No	80
Emergency Action Code	2X

Regulations concerning the International Carriage of Dangerous Goods by Rail (RID)Additional information

Classification code	C1
Danger label(s)	8



Excepted quantities (EQ)	E1
Limited quantities (LQ)	5 L
Transport category (TC)	3
Hazard identification No	80

International Maritime Dangerous Goods Code (IMDG) - Additional information

Proper shipping name	FERRIC CHLORIDE SOLUTION
Particulars in the shipper's declaration	UN2582, FERRIC CHLORIDE SOLUTION, 8, III
Marine pollutant	-
Danger label(s)	8



Special provisions (SP)	223
Excepted quantities (EQ)	E1
Limited quantities (LQ)	5 L
EmS	F-A, S-B
Stowage category	A
Segregation group	1 - Acids

International Civil Aviation Organization (ICAO-IATA/DGR) - Additional information

Proper shipping name	Ferric chloride solution
Particulars in the shipper's declaration	UN2582, Ferric chloride solution, 8, III
Danger label(s)	8



Special provisions (SP)	A3
Excepted quantities (EQ)	E1
Limited quantities (LQ)	1 L

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SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Relevant provisions of the European Union (EU)

Seveso Directive

2012/18/EU (Seveso III)			
No	Dangerous substance/hazard categories	Qualifying quantity (tonnes) for the application of lower and upper-tier requirements	Notes
	not assigned		

Deco-Paint Directive

VOC content	0 %
VOC content (Water content was discounted)	0 g/l

Industrial Emissions Directive (IED)

VOC content	0 %
VOC content (Water content was discounted)	0 g/l

Directive on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS)

none of the ingredients are listed

Regulation concerning the establishment of a European Pollutant Release and Transfer Register (PRTR)

none of the ingredients are listed

Water Framework Directive (WFD)

List of pollutants (WFD)				
Name of substance	Name acc. to inventory	CAS No	Listed in	Remarks
Iron(III) chloride	Metals and their compounds		a)	

Legend

A) Indicative list of the main pollutants

Regulation on the marketing and use of explosives precursors

none of the ingredients are listed

Regulation on drug precursors

Name of substance	CAS No	Wt%	Classification	CN Code	Threshold level
Hydrochloric acid %	7647-01-0	1	Category 3	2806 10 00	

Regulation on substances that deplete the ozone layer (ODS)

none of the ingredients are listed

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Regulation concerning the export and import of hazardous chemicals (PIC)

none of the ingredients are listed

Regulation on persistent organic pollutants (POP)

none of the ingredients are listed

Other information

Directive 94/33/EC on the protection of young people at work. Observe employment restrictions under the Maternity Protection Directive (92/85/EEC) for expectant or nursing mothers.

UN Convention against Illicit Traffic in Narcotic Drugs and Psychotropic Substances

Name of substance	CAS No	Listed in	HS code
Hydrochloric acid %	7647-01-0	Table II	2806.10

National inventories

Country	Inventory	Status
AU	AIIC	all ingredients are listed
CA	DSL	all ingredients are listed
CN	IECSC	all ingredients are listed
EU	ECSI	all ingredients are listed
EU	REACH Reg.	all ingredients are listed
JP	CSCL-ENCS	all ingredients are listed
KR	KECI	all ingredients are listed
MX	INSQ	all ingredients are listed
NZ	NZIoC	all ingredients are listed
PH	PICCS	all ingredients are listed
TR	CICR	not all ingredients are listed
TW	TCSI	all ingredients are listed
US	TSCA	all ingredients are listed as "ACTIVE"

Legend

AIIC	Australian Inventory of Industrial Chemicals
CICR	Chemical Inventory and Control Regulation
CSCL-ENCS	List of Existing and New Chemical Substances (CSCL-ENCS)
DSL	Domestic Substances List (DSL)
ECSI	EC Substance Inventory (EINECS, ELINCS, NLP)
IECSC	Inventory of Existing Chemical Substances Produced or Imported in China
INSQ	National Inventory of Chemical Substances
KECI	Korea Existing Chemicals Inventory
NZIoC	New Zealand Inventory of Chemicals
PICCS	Philippine Inventory of Chemicals and Chemical Substances (PICCS)
REACH Reg.	REACH registered substances
TCSI	Taiwan Chemical Substance Inventory
TSCA	Toxic Substance Control Act

15.2 Chemical Safety Assessment

Chemical safety assessments for substances in this mixture were not carried out.

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SECTION 16: Other information

Indication of changes (revised safety data sheet)

Section	Former entry (text/value)	Actual entry (text/value)	Safety-relevant
2.2	Labelling of packages where the contents do not exceed 125 ml: Signal word: Danger		yes
2.2		Labelling of packages where the contents do not exceed 125 ml: change in the listing (table)	yes
2.2		Labelling of packages where the contents do not exceed 125 ml: change in the listing (table)	yes
2.2		Labelling of packages where the contents do not exceed 125 ml: change in the listing (table)	yes
2.2	contains: Iron(III) chloride, Hydrochloric acid %		yes
2.3	Results of PBT and vPvB assessment: This mixture does not contain any substances that are assessed to be a PBT or a vPvB.	Results of PBT and vPvB assessment: Does not contain a PBT-/vPvB-substance in a concentration of $\geq 0,1\%$.	yes
2.3		Endocrine disrupting properties: Does not contain an endocrine disruptor (EDC) in a concentration of $\geq 0,1\%$.	yes
14.8		Regulations concerning the International Carriage of Dangerous Goods by Rail (RID)Additional information	yes
14.8		Classification code: C1	yes
14.8		Danger label(s): 8	yes
14.8		Danger label(s): change in the listing (table)	yes
14.8		Excepted quantities (EQ): E1	yes
14.8		Limited quantities (LQ): 5 L	yes
14.8		Transport category (TC): 3	yes
14.8		Hazard identification No: 80	yes
15.1	Restrictions according to REACH, Annex XVII		yes
15.1		Dangerous substances with restrictions (REACH, Annex XVII): change in the listing (table)	yes
15.1	List of substances subject to authorisation (REACH, Annex XIV)/SVHC - candidate list: None of the ingredients are listed.		yes

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Section	Former entry (text/value)	Actual entry (text/value)	Safety-relevant
15.1	VOC content: 0 % , 0 g/l	VOC content: 0 %	yes
15.1		VOC content (Water content was discounted): 0 g/l	yes
15.1		Regulation on drug precursors: change in the listing (table)	yes
15.1		National inventories: change in the listing (table)	yes

Abbreviations and acronyms

Abbr.	Descriptions of used abbreviations
2000/39/EC	Commission Directive establishing a first list of indicative occupational exposure limit values in implementation of Council Directive 98/24/EC
Acute Tox.	Acute toxicity
ADR	Accord relatif au transport international des marchandises dangereuses par route (Agreement concerning the International Carriage of Dangerous Goods by Road)
ATE	Acute Toxicity Estimate
BCF	Bioconcentration factor
BOD	Biochemical Oxygen Demand
CAS	Chemical Abstracts Service (service that maintains the most comprehensive list of chemical substances)
Ceiling-C	Ceiling value
CN Code	Combined Nomenclature
COD	Chemical oxygen demand
DGR	Dangerous Goods Regulations (see IATA/DGR)
DNEL	Derived No-Effect Level
EC No	The EC Inventory (EINECS, ELINCS and the NLP-list) is the source for the seven-digit EC number, an identifier of substances commercially available within the EU (European Union)
EH40/2005	EH40/2005 Workplace exposure limits (http://www.nationalarchives.gov.uk/doc/open-government-licence/)
EINECS	European Inventory of Existing Commercial Chemical Substances
ELINCS	European List of Notified Chemical Substances
EmS	Emergency Schedule
Eye Dam.	Seriously damaging to the eye
Eye Irrit.	Irritant to the eye
GHS	"Globally Harmonized System of Classification and Labelling of Chemicals" developed by the United Nations
HS	Harmonized Commodity Description and Coding System (Harmonized System, drawn up by the World Customs Organisation)
IATA	International Air Transport Association

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Abbr.	Descriptions of used abbreviations
IATA/DGR	Dangerous Goods Regulations (DGR) for the air transport (IATA)
ICAO	International Civil Aviation Organization
ICAO-TI	Technical instructions for the safe transport of dangerous goods by air
IMDG	International Maritime Dangerous Goods Code
IMDG-Code	International Maritime Dangerous Goods Code
index No	The Index number is the identification code given to the substance in Part 3 of Annex VI to Regulation (EC) No 1272/2008
IOELV	Indicative occupational exposure limit value
LD50	Lethal Dose 50 %: the LD50 corresponds to the dose of a tested substance causing 50 % lethality during a specified time interval
log KOW	n-Octanol/water
Met. Corr.	Substance or mixture corrosive to metals
NLP	No-Longer Polymer
PBT	Persistent, Bioaccumulative and Toxic
ppm	Parts per million
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals
RID	Règlement concernant le transport International ferroviaire des marchandises Dangereuses (Regulations concerning the International carriage of Dangerous goods by Rail)
Skin Corr.	Corrosive to skin
Skin Irrit.	Irritant to skin
Skin Sens.	Skin sensitisation
STEL	Short-term exposure limit
STOT SE	Specific target organ toxicity - single exposure
TWA	Time-weighted average
VOC	Volatile Organic Compounds
vPvB	Very Persistent and very Bioaccumulative
WEL	Workplace exposure limit

Key literature references and sources for data

Agreement concerning the International Carriage of Dangerous Goods by Road (ADR). Regulations concerning the International Carriage of Dangerous Goods by Rail (RID). International Maritime Dangerous Goods Code (IMDG). Dangerous Goods Regulations (DGR) for the air transport (IATA).

Classification procedure

Physical and chemical properties. The classification is based on tested mixture.
Health hazards. Environmental hazards. The method for classification of the mixture is based on ingredients of the mixture (additivity formula).

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List of relevant phrases (code and full text as stated in section 2 and 3)

Code	Text
H290	May be corrosive to metals.
H302	Harmful if swallowed.
H314	Causes severe skin burns and eye damage.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H318	Causes serious eye damage.
H335	May cause respiratory irritation.

Disclaimer

This information is based upon the present state of our knowledge. This SDS has been compiled and is solely intended for this product.